Aditya Surendra Rekhe

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Portfolio | GitHub | LinkedIn

EDUCATION

Bachelor of Engineering, Computer Engineering Oct 2021 – Jul 2025
Dr. D. Y. Patil Institute of Technology, Savitribai Phule Pune University, Pune CGPA 8.68 / 10

Higher Secondary Certificate (HSC), Maharashtra State Board May 2021
P. S. Deshmukh English Medium School and Junior College, Shahapur 89 %

Secondary School Certificate (SSC), Maharashtra State Board March 2019
P. S. Deshmukh English Medium School and Junior College, Shahapur 85.6 %

PUBLICATIONS

Differential Security Analysis of EU Digital Identity Credential Libraries 2026
Sole-authored preprint, Zenodo DOI | Code

- To the best of my knowledge, the first systematic differential and adversarial evaluation of the libraries implementing the two credential formats mandated for the EU Digital Identity Wallet under eIDAS 2.0.
- Developed 30 clause-based validation rules and 40 adversarial test cases, run against 7 implementations across 6 codebases in three programming languages (122 evaluations).
- Found implementation-specific security and standards-compliance issues and prepared findings for responsible disclosure.

TruthLens: A Verification-Gated Pipeline for Evidence-Grounded Fact-Checking of Short-Form Political Video 2026
Sole-authored preprint, Zenodo DOI | Code

- Built a nine-stage pipeline that retrieves sources, verifies claims, and requires supporting evidence before producing a verdict.
- Defined *support validity* and showed it is separable from label accuracy: validator recall improved from 16.7 % to 40 % at 100 % precision.
- Developed a 202-item benchmark based on professionally fact-checked short-form videos and evaluated the system using a separate validation layer.

EXPERIENCE

Independent Researcher May 2026 – Present
Self-directed research, Pune, India

- Designed and built adversarial evaluation infrastructure for cryptographic credential verification and for retrieval-based fact-checking systems.
- Ran coordinated vulnerability disclosure with open-source maintainers; prepared a specification-level submission for the relevant standards body.

Software Engineer Sep 2025 – Apr 2026
Stratacent Technologies Pvt. Ltd., Pune, India

- Handled incident management across database, cybersecurity and infrastructure domains in production environments.
- Performed root cause analysis for critical system failures and operational incidents.
- Contributed to development of a GenAI-powered internal chatbot system.
- Worked with cross-functional teams to improve system reliability and operational workflows.

PROJECTS

Aheadly: Student Mentorship Marketplace GitHub | Website

Next.js 16 (React 19) · TypeScript · PostgreSQL · Drizzle ORM · Tailwind CSS v4 · Vitest & Playwright · GitHub Actions

- Designed a two-sided marketplace connecting students with mentors for career and study-abroad guidance, covering market research, system architecture, database and API design.
- Built 580 automated tests across unit, real-Postgres integration and multi-browser E2E testing, with GitHub Actions, CodeQL and dependency scanning; secured the platform using STRIDE, OWASP Top 10 / API Top 10 and ASVS Level 2, with verification, fraud prevention and dispute handling.

NeuralCraft: Interactive 3D Playground for Learning AI/ML GitHub | Website

TypeScript, Vite, Three.js, Blockly, CodeMirror 6, Pyodide, Web Audio API, Vitest

- Built a browser-based 3D voxel-world playground teaching AI, ML and data science through 36 lessons and 6 interactive labs (gradient descent, k-means, neural networks, embeddings, Q-learning).
- Implemented a code sandbox (Blockly, Python, JavaScript), a 3D environment with dynamic lighting and A* pathfinding, and a local-first XP/badge system, deployed free with no backend.

CareCircle: Community Platform for Chronic Illness GitHub | Website

React 19, TypeScript, Tailwind CSS, Zustand, TanStack Query, Supabase (Auth, Postgres/RLS, Realtime)

- Built a community platform unifying forum discussion, real-time chat, Q&A and condition tracking for people managing chronic illness, with dedicated communities per condition and support for comorbidities.
- Implemented authentication, row-level security and realtime features on Supabase, deployed serverless via GitHub Pages, with CI/CD running lint, test, build and migration checks on every PR.

TECHNICAL SKILLS

Languages	Python, C, C++, Java, JavaScript / TypeScript, Solidity, SQL
Security	Applied cryptography; smart contracts; digital forensics
ML & Data	Machine learning, deep learning, NLP; retrieval-augmented generation; LangChain; LLM orchestration
Frameworks	React.js, Node.js, FastAPI, Next.js, MongoDB, PostgreSQL, pgvector, IPFS, WebRTC, REST API design

ACHIEVEMENTS

- **Blockchain Lead**, *Google Developer Groups (GDG) student chapter* Mar 2023 – Feb 2025
- **Guest Lecturer**, *blockchain and dApp development workshops at multiple universities*
- **Third Prize**, *intercollegiate blockchain hackathon, All India Blockchain Alliance* Sep 2023
- **Selected Participant**, *Unfold 2023, among India's largest blockchain hackathons* Dec 2023